

DBMS PRACTICE 3

ORDERS TABLE

STEP 1: CREATE TABLE

```
MySQL 8.0 Command Line Cli x + v

mysql> use task3;
Database changed
mysql> create table orders (
  -> ord_no int primary key,
  -> purch_amt decimal(10,2),
  -> ord_date date,
  -> customer_id int,
  -> salesman_id int
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> desc orders;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| ord_no | int | NO | PRI | NULL | |
| purch_amt | decimal(10,2) | YES | | NULL | |
| ord_date | date | YES | | NULL | |
| customer_id | int | YES | | NULL | |
| salesman_id | int | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.03 sec)

mysql>
```

STEP 2: INSERT DATA

```
MySQL 8.0 Command Line Cli x + v

5 rows in set (0.03 sec)

mysql> insert into orders values
  -> (70001,150.50,'2012-10-05',3005,5002),
  -> (70009,270.65,'2012-09-10',3001,5005),
  -> (70002,65.26,'2012-10-05',3002,5001),
  -> (70004,110.50,'2012-08-17',3009,5003),
  -> (70007,948.50,'2012-09-10',3005,5002),
  -> (70005,2400.60,'2012-07-27',3007,5001),
  -> (70008,5760.00,'2012-09-10',3002,5001),
  -> (70010,1983.43,'2012-10-10',3004,5006),
  -> (70003,2480.40,'2012-10-10',3009,5003),
  -> (70012,250.45,'2012-06-27',3008,5002),
  -> (70011,75.29,'2012-08-17',3003,5007),
  -> (70013,3045.60,'2012-04-25',3002,5001);
Query OK, 12 rows affected (0.03 sec)
Records: 12 Duplicates: 0 Warnings: 0

mysql> select * from orders;
+-----+-----+-----+-----+-----+
| ord_no | purch_amt | ord_date | customer_id | salesman_id |
+-----+-----+-----+-----+-----+
| 70001 | 150.50 | 2012-10-05 | 3005 | 5002 |
| 70002 | 65.26 | 2012-10-05 | 3002 | 5001 |
| 70003 | 2480.40 | 2012-10-10 | 3009 | 5003 |
| 70004 | 110.50 | 2012-08-17 | 3009 | 5003 |
| 70005 | 2400.60 | 2012-07-27 | 3007 | 5001 |
| 70007 | 948.50 | 2012-09-10 | 3005 | 5002 |
| 70008 | 5760.00 | 2012-09-10 | 3002 | 5001 |
| 70009 | 270.65 | 2012-09-10 | 3001 | 5005 |
| 70010 | 1983.43 | 2012-10-10 | 3004 | 5006 |
| 70011 | 75.29 | 2012-08-17 | 3003 | 5007 |
| 70012 | 250.45 | 2012-06-27 | 3008 | 5002 |
| 70013 | 3045.60 | 2012-04-25 | 3002 | 5001 |
+-----+-----+-----+-----+-----+
12 rows in set (0.00 sec)
```

QUESTION 1 - WHERE CLAUSE

```
mysql> select * from orders
-> where purch_amt > 2000;
+-----+-----+-----+-----+-----+
| ord_no | purch_amt | ord_date | customer_id | salesman_id |
+-----+-----+-----+-----+-----+
| 70003 | 2480.40 | 2012-10-10 | 3009 | 5003 |
| 70005 | 2400.60 | 2012-07-27 | 3007 | 5001 |
| 70008 | 5760.00 | 2012-09-10 | 3002 | 5001 |
| 70013 | 3045.60 | 2012-04-25 | 3002 | 5001 |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> |
```

QUESTION 2 - WHERE CLAUSE

```
mysql> select * from orders
-> where ord_date = '2012-09-10';
+-----+-----+-----+-----+-----+
| ord_no | purch_amt | ord_date | customer_id | salesman_id |
+-----+-----+-----+-----+-----+
| 70007 | 948.50 | 2012-09-10 | 3005 | 5002 |
| 70008 | 5760.00 | 2012-09-10 | 3002 | 5001 |
| 70009 | 270.65 | 2012-09-10 | 3001 | 5005 |
+-----+-----+-----+-----+-----+
3 rows in set (0.02 sec)

mysql> |
```

QUESTION 3 - WHERE CLAUSE

```
mysql> select * from orders
-> where salesman_id = 5001;
+-----+-----+-----+-----+-----+
| ord_no | purch_amt | ord_date | customer_id | salesman_id |
+-----+-----+-----+-----+-----+
| 70002 | 65.26 | 2012-10-05 | 3002 | 5001 |
| 70005 | 2400.60 | 2012-07-27 | 3007 | 5001 |
| 70008 | 5760.00 | 2012-09-10 | 3002 | 5001 |
| 70013 | 3045.60 | 2012-04-25 | 3002 | 5001 |
+-----+-----+-----+-----+-----+
4 rows in set (0.02 sec)

mysql> |
```

QUESTION 4 - ORDER BY

```
mysql> select * from orders
-> order by purch_amt desc;
+-----+-----+-----+-----+-----+
| ord_no | purch_amt | ord_date | customer_id | salesman_id |
+-----+-----+-----+-----+-----+
| 70008 | 5760.00 | 2012-09-10 | 3002 | 5001 |
| 70013 | 3045.60 | 2012-04-25 | 3002 | 5001 |
| 70003 | 2480.40 | 2012-10-10 | 3009 | 5003 |
| 70005 | 2400.60 | 2012-07-27 | 3007 | 5001 |
| 70010 | 1983.43 | 2012-10-10 | 3004 | 5006 |
| 70007 | 948.50 | 2012-09-10 | 3005 | 5002 |
| 70009 | 270.65 | 2012-09-10 | 3001 | 5005 |
| 70012 | 250.45 | 2012-06-27 | 3008 | 5002 |
| 70001 | 150.50 | 2012-10-05 | 3005 | 5002 |
| 70004 | 110.50 | 2012-08-17 | 3009 | 5003 |
| 70011 | 75.29 | 2012-08-17 | 3003 | 5007 |
| 70002 | 65.26 | 2012-10-05 | 3002 | 5001 |
+-----+-----+-----+-----+-----+
12 rows in set (0.00 sec)

mysql> |
```

QUESTION 5 - ORDER BY

```
mysql> select * from orders
-> order by ord_date;
+-----+-----+-----+-----+-----+
| ord_no | purch_amt | ord_date | customer_id | salesman_id |
+-----+-----+-----+-----+-----+
| 70013 | 3045.60 | 2012-04-25 | 3002 | 5001 |
| 70012 | 250.45 | 2012-06-27 | 3008 | 5002 |
| 70005 | 2400.60 | 2012-07-27 | 3007 | 5001 |
| 70004 | 110.50 | 2012-08-17 | 3009 | 5003 |
| 70011 | 75.29 | 2012-08-17 | 3003 | 5007 |
| 70007 | 948.50 | 2012-09-10 | 3005 | 5002 |
| 70008 | 5760.00 | 2012-09-10 | 3002 | 5001 |
| 70009 | 270.65 | 2012-09-10 | 3001 | 5005 |
| 70001 | 150.50 | 2012-10-05 | 3005 | 5002 |
| 70002 | 65.26 | 2012-10-05 | 3002 | 5001 |
| 70003 | 2480.40 | 2012-10-10 | 3009 | 5003 |
| 70010 | 1983.43 | 2012-10-10 | 3004 | 5006 |
+-----+-----+-----+-----+-----+
12 rows in set (0.00 sec)

mysql> |
```

QUESTION 6 - SUM()

```
mysql> select sum(purch_amt) as total_revenue
-> from orders;
+-----+
| total_revenue |
+-----+
| 17541.18 |
+-----+
1 row in set (0.00 sec)

mysql> |
```

QUESTION 7 - AVG()

```
mysql> select sum(purch_amt) as total_revenue
-> from orders;
+-----+
| total_revenue |
+-----+
| 17541.18 |
+-----+
1 row in set (0.00 sec)

mysql> select avg(purch_amt) as average_order
-> from orders;
+-----+
| average_order |
+-----+
| 1461.765000 |
+-----+
1 row in set (0.00 sec)

mysql> |
```

QUESTION 8 - MAX()

```
mysql> select sum(purch_amt) as total_revenue
-> from orders;
+-----+
| total_revenue |
+-----+
|      17541.18 |
+-----+
1 row in set (0.00 sec)

mysql> select avg(purch_amt) as average_order
-> from orders;
+-----+
| average_order |
+-----+
|    1461.765000 |
+-----+
1 row in set (0.00 sec)

mysql> select max(purch_amt) as highest_order
-> from orders;
+-----+
| highest_order |
+-----+
|         5760.00 |
+-----+
1 row in set (0.00 sec)

mysql> |
```

QUESTION 9 - MIN()

```
mysql> select avg(purch_amt) as average_order
-> from orders;
+-----+
| average_order |
+-----+
|    1461.765000 |
+-----+
1 row in set (0.00 sec)

mysql> select max(purch_amt) as highest_order
-> from orders;
+-----+
| highest_order |
+-----+
|         5760.00 |
+-----+
1 row in set (0.00 sec)

mysql> select min(purch_amt) as lowest_order
-> from orders;
+-----+
| lowest_order |
+-----+
|          65.26 |
+-----+
1 row in set (0.00 sec)

mysql> |
```

QUESTION 10 - COUNT()

```
mysql> select avg(purch_amt) as average_order
-> from orders;
+-----+
| average_order |
+-----+
|    1461.765000 |
+-----+
1 row in set (0.00 sec)

mysql> select max(purch_amt) as highest_order
-> from orders;
+-----+
| highest_order |
+-----+
|         5760.00 |
+-----+
1 row in set (0.00 sec)

mysql> select min(purch_amt) as lowest_order
-> from orders;
+-----+
| lowest_order |
+-----+
|          65.26 |
+-----+
1 row in set (0.00 sec)

mysql> select count(*) as total_orders
-> from orders;
+-----+
| total_orders |
+-----+
|          12 |
+-----+
1 row in set (0.00 sec)
```

QUESTION 11 - GROUP BY

```
mysql> select salesman_id, sum(purch_amt) as total_sales
-> from orders
-> group by salesman_id;
+-----+-----+
| salesman_id | total_sales |
+-----+-----+
|          5002 |      1349.45 |
|          5001 |     11271.46 |
|          5003 |      2590.90 |
|          5005 |       270.65 |
|          5006 |     1983.43 |
|          5007 |        75.29 |
+-----+-----+
6 rows in set (0.00 sec)

mysql> |
```

QUESTION 12 - GROUP BY

```
mysql> select customer_id, sum(purch_amt) as total_purchase
-> from orders
-> group by customer_id;
+-----+-----+
| customer_id | total_purchase |
+-----+-----+
|          3005 |      1099.00 |
|          3002 |     8870.86 |
|          3009 |      2590.90 |
|          3007 |      2400.60 |
|          3001 |       270.65 |
|          3004 |     1983.43 |
|          3003 |        75.29 |
|          3008 |       250.45 |
+-----+-----+
8 rows in set (0.00 sec)

mysql> |
```

QUESTION 13 - GROUP BY + MAX()

```
mysql> select customer_id, max(purch_amt) as highest_purchase
-> from orders
-> group by customer_id;
+-----+-----+
| customer_id | highest_purchase |
+-----+-----+
|          3005 |          948.50 |
|          3002 |         5760.00 |
|          3009 |         2480.40 |
|          3007 |         2400.60 |
|          3001 |          270.65 |
|          3004 |         1983.43 |
|          3003 |           75.29 |
|          3008 |          250.45 |
+-----+-----+
8 rows in set (0.00 sec)

mysql> |
```

QUESTION 14 - GROUP BY + HAVING

```
mysql> select salesman_id, sum(purch_amt) as total_sales
-> from orders
-> group by salesman_id
-> having sum(purch_amt) > 3000;
+-----+-----+
| salesman_id | total_sales |
+-----+-----+
|          5001 |     11271.46 |
+-----+-----+
1 row in set (0.00 sec)

mysql> |
```

QUESTION 15 - GROUP BY + HAVING

```
mysql> select customer_id, sum(purch_amt) as total_purchases
-> from orders
-> group by customer_id
-> having sum(purch_amt) > 2500;
+-----+-----+
| customer_id | total_purchases |
+-----+-----+
|          3002 |          8870.86 |
|          3009 |          2590.90 |
+-----+-----+
2 rows in set (0.02 sec)

mysql> |
```

QUESTION 16 - GROUP BY + HAVING

```
mysql> select salesman_id,
-> count(*) as total_orders
-> from orders
-> group by salesman_id
-> having count(*) >= 2;
+-----+-----+
| salesman_id | total_orders |
+-----+-----+
|          5002 |           3 |
|          5001 |           4 |
|          5003 |           2 |
+-----+-----+
3 rows in set (0.02 sec)

mysql> |
```

QUESTION 17 - GROUP BY + HAVING + ORDER BY

```
mysql> select customer_id,
-> sum(purch_amt) as total_purchase
-> from orders
-> group by customer_id
-> having sum(purch_amt) > 1000
-> order by total_purchase desc;
+-----+-----+
| customer_id | total_purchase |
+-----+-----+
|          3002 |          8870.86 |
|          3009 |          2590.90 |
|          3007 |          2400.60 |
|          3004 |          1983.43 |
|          3005 |          1099.00 |
+-----+-----+
5 rows in set (0.00 sec)

mysql> |
```

QUESTION 18 - BETWEEN + HAVING

```
mysql> select customer_id,
-> max(purch_amt) as max_purchase
-> from orders
-> group by customer_id
-> having max(purch_amt) between 2000 and 6000;
+-----+-----+
| customer_id | max_purchase |
+-----+-----+
|          3002 |          5760.00 |
|          3009 |          2480.40 |
|          3007 |          2400.60 |
+-----+-----+
3 rows in set (0.00 sec)

mysql> |
```

QUESTION 19 - COUNT + HAVING

```
mysql> select salesman_id,  
-> count(*) as total_orders  
-> from orders  
-> group by salesman_id  
-> having count(*) >= 2;  
+-----+-----+  
| salesman_id | total_orders |  
+-----+-----+  
| 5002 | 3 |  
| 5001 | 4 |  
| 5003 | 2 |  
+-----+-----+  
3 rows in set (0.00 sec)  
mysql> |
```

QUESTION 20 - MAX + GROUP BY + HAVING

```
mysql> select ord_date,  
-> max(purch_amt) as highest_purchase  
-> from orders  
-> group by ord_date  
-> having max(purch_amt) > 2000;  
+-----+-----+  
| ord_date | highest_purchase |  
+-----+-----+  
| 2012-10-10 | 2480.40 |  
| 2012-07-27 | 2400.60 |  
| 2012-09-10 | 5760.00 |  
| 2012-04-25 | 3045.60 |  
+-----+-----+  
4 rows in set (0.02 sec)  
mysql> |
```